

SQL Technical Report: Dunder Mifflin

Emma Liu, Simon Wang, Hugo Berthoumieux, Thu Nguyen

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EVALUATING THE ACQUISITION OF DUNDER MIFFLIN

I. Introduction

- **Background:** In January 2017, as a data analyst at Dunder Mifflin, we were tasked by Michael Scott to perform an exploratory data analysis to support the leadership team's decision on acquiring the smaller paper company, Parch and Posey.
- **Objective:** To assess Parch and Posey's current business performance, analyze its revenue streams, growth potential, sales efficiency, and marketing strategies, providing insights to inform Dunder Mifflin's acquisition decision.

II. Data Description

- **Data Source:** The analysis covers data from Parch and Posey's operations across multiple regions (Northeast, Midwest, Southeast, West, International, South, and North), involving a range of products, including standard, glossy and poster paper. The company serves a broad customer base of 352 accounts, supported by a team of 52 sales representatives, generating over \$23.14 million in revenue.
- **Data Structure:** The data includes product sales, account details, representative assignments by region, and customer metrics such as repeat customer and unique monthly counts. This setup allows a breakdown of sales by product type and region, alongside sales rep distribution to optimize coverage and revenue generation.

III. Methodology

The methodology involved querying and analyzing various metrics, such as monthly revenue growth (2014-2016), customer trends, and order growth by product type. Revenue and customer metrics were used to identify sales patterns and regional performance, while SQL queries helped map each sales rep's account allocation and regional presence.

Q1: What is the current state of the company?

a) Size of customer base
select count(distinct name)
from accounts a

Number of employees (representative)
select count(*)
from sales_reps *sr*

b) Regions where the company operates

select *
from region

c) Number of representatives per region

select r.name, count(*)
from sales_reps *sr* left join region *r* on *sr*.region_id = *r*.id
group by r.name

Q2: Look into revenue streams

select sum(standard_qty)
from orders o

select sum(gloss_qty)
from orders o

select sum(poster_qty)
from orders o

select avg(standard_qty) as stavg, avg(gloss_qty) as glossavg, avg(poster_qty) as
posavg from orders o

select sum(standard_amt_usd), sum(gloss_amt_usd), sum(poster_amt_usd)
from orders o

select sum(total)
from orders o

select sum(total_amt_usd)
from orders o

select sum(standard_amt_usd)/sum(standard_qty), sum(gloss_amt_usd)/sum(gloss_qty),
sum(poster_amt_usd)/sum(poster_qty)
from orders o

Q3: Is the business growing?

a) Revenue growth over time

SELECT DATE_TRUNC('month', occurred_at) as month,
SUM(standard_qty) as total_standard, SUM(poster_qty) as total_poster, SUM(orders.gloss_qty)
as total_glossy, ROUND(AVG(standard_qty), 2) as avg_standard_per_order,
ROUND(AVG(poster_qty), 2) as avg_poster_per_order,
ROUND(AVG(orders.gloss_qty), 2) as avg_glossy_per_order
FROM orders

```
GROUP BY 1  
ORDER BY 1;
```

b) Orders growth by type of product

```
SELECT a.id as account_id, a.name as account_name,  
COUNT(o.id) as number_of_orders, SUM(o.total_amt_usd) as total_revenue,  
SUM(o.total_amt_usd)/COUNT(o.id) as avg_revenue_per_order FROM accounts a  
LEFT JOIN orders o ON a.id = o.account_id  
GROUP BY 1, 2  
ORDER BY total_revenue DESC;
```

To complement our analysis, we aimed to provide a more global view of sales performance across the three main product types (standard, poster, and glossy) over a span of three years (2014, 2015, and 2016). This broader perspective helps us understand not only monthly trends but also the cumulative growth and relative popularity of each product type over a longer period.

```
SELECT  
    DATE_TRUNC('year', occurred_at) AS year,  
    SUM(standard_amt_usd) AS total_standard_sales,  
    SUM(poster_amt_usd) AS total_poster_sales,  
    SUM(glossy_amt_usd) AS total_glossy_sales  
FROM orders  
WHERE occurred_at BETWEEN '2014-01-01' AND '2016-12-31'  
GROUP BY year  
ORDER by year;
```

c) Customer growth over time

```
WITH monthly_orders AS (  
    SELECT account_id, DATE_TRUNC('month', occurred_at) AS month, COUNT(id) AS  
    order_count  
    FROM orders  
    GROUP BY account_id, DATE_TRUNC('month', occurred_at)  
)  
SELECT month, COUNT(DISTINCT account_id) AS total_customers, COUNT(DISTINCT  
CASE WHEN order_count > 1 THEN account_id END) AS repeat_customers  
FROM monthly_orders  
GROUP BY month  
ORDER BY month;
```

Q4: Could we increase sales efficiency?

```

SELECT sr.id AS sales_rep_id,
       sr.name AS sales_rep_name,
       r.id AS region_id,
       r.name AS region_name,
       COUNT(a.id) AS num_accounts
FROM sales_reps sr
JOIN region r ON r.id = sr.region_id
JOIN accounts a ON a.sales_rep_id = sr.id
GROUP BY sr.id, sr.name, r.id, r.name
ORDER BY num_accounts DESC;

```

```

SELECT sr.id AS sales_rep_id,
       sr.name AS sales_rep_name,
       r.id AS region_id,
       r.name AS region_name,
       COUNT(DISTINCT a.id) AS num_accounts,
       SUM(o.total) AS total_sales
FROM sales_reps sr
JOIN region r ON sr.region_id = r.id
LEFT JOIN accounts a ON sr.id = a.sales_rep_id
LEFT JOIN orders o ON a.id = o.account_id
GROUP BY sr.id, sr.name, r.id, r.name
ORDER BY total_sales DESC;

```

```

select * from sales_reps sr
right join region r
on r.id = sr.region_id

```

```

SELECT r.id AS region_id, r.name AS region_name, COUNT(sr.name) AS num_sales_reps
FROM sales_reps sr
RIGHT JOIN region r ON r.id = sr.region_id
GROUP BY r.id, r.name
ORDER BY num_sales_reps DESC;

```

```

SELECT r.id AS region_id,
       r.name AS region_name,
       SUM(o.total) AS total_sales
FROM region r
JOIN sales_reps sr ON sr.region_id = r.id
JOIN accounts a ON a.sales_rep_id = sr.id
JOIN orders o ON a.id = o.account_id
GROUP BY r.id, r.name
ORDER BY total_sales DESC;

```

Q5: Marketing Strategy

a) Food industry

```

SELECT name AS account_name, 'Food' AS industry
FROM accounts
WHERE name LIKE '%Food%';

```

b) Health industry

```

SELECT name AS account_name, 'Health' AS industry
FROM accounts
WHERE name LIKE '%Health%' OR name LIKE '%Hospital%';

```

c) Retail/ Consumer Goods industry

```
SELECT name AS account_name, 'Retail/Consumer Goods' AS industry
FROM accounts
WHERE name LIKE '%Store%' OR name LIKE '%Shop%' OR name LIKE '%Retail%' OR name
LIKE '%Mart%';
```

d) Marketing/ Advertising Industry

```
SELECT name AS account_name, 'Marketing/Advertising' AS industry
FROM accounts
WHERE name LIKE '%Agency%' OR name LIKE '%Marketing%' OR name LIKE '%Ad%' OR
name LIKE '%Design%';
```

e) Education industry

```
SELECT name AS account_name, 'Education' AS industry
FROM accounts
WHERE name LIKE '%School%' OR name LIKE '%University%' OR name LIKE '%College%'
OR name LIKE '%Education%';
```

f) Publishing/ Printing

```
SELECT name AS account_name, 'Publishing/Printing' AS industry
FROM accounts
WHERE name LIKE '%Publish%' OR name LIKE '%Print%' OR name LIKE '%Press%' OR
name LIKE '%Media%';
```

g) Least Used Channel

```
SELECT region, channel AS least_used_channel
FROM (
  SELECT
    r.name AS region,
    w.channel,
    COUNT(w.id) AS web_event_count,
    RANK() OVER (PARTITION BY r.name ORDER BY COUNT(w.id) ASC) AS rank
  FROM web_events w
  JOIN accounts a ON w.account_id = a.id
  JOIN sales_reps s ON a.sales_rep_id = s.id
  JOIN region r ON s.region_id = r.id
  GROUP BY r.name, w.channel
) AS ranked_channels
WHERE rank = 1;
```

IV. Results

Q1: What is the current state of the company?

The company currently operates across a broad geographic footprint, including the Northeast, Midwest, Southeast, West, International, South, and North regions, reflecting a significant market reach. It offers a diverse range of products, including standard paper, glossy paper, and poster paper, catering to various customer needs and preferences. With a robust customer base of 352 accounts, the company has generated over \$23.14 million in revenue. Supported by a team of 52 dedicated sales representatives, the company has achieved a total sales volume of 3.68 million units, indicating a strong position within its industry and a substantial demand for its products across its wide operational landscape.

a) Size of customer base

The screenshot shows a database query window titled "Results 1". The SQL query entered is `select count(distinct name)`. The result is displayed in a grid with two columns: an index column and a column labeled "0-9 count". The first row shows the value 352.

	0-9 count
1	352

b) Number of employees (representative)

The screenshot shows a database query window titled "Results 1". The SQL query entered is `select count(*) from sales`. The result is displayed in a grid with two columns: an index column and a column labeled "0-9 count". The first row shows the value 52.

	0-9 count
1	52

c) Regions where the company operates

The screenshot shows a database query window titled "region 1". The SQL query entered is `select * from region`. The result is displayed in a grid with three columns: an index column, a column labeled "0-9 id", and a column labeled "A-Z name". The regions listed are Northeast, Midwest, Southeast, West, International, South, and North.

	0-9 id	A-Z name
1	1	Northeast
2	2	Midwest
3	3	Southeast
4	4	West
5	5	International
6	6	South
7	7	North

d) Number of representatives per region

region 1 X

select r.name, count(*) fr

	A-Z name	0-9 count
1	[NULL]	1
2	Midwest	9
3	Southeast	10
4	West	10
5	International	1
6	Northeast	21



The chart shows that the Northeast region has the highest number of sales representatives, indicating a strong focus there, likely due to high demand or market opportunity. The Southeast, West, and Midwest have moderate representation, while the international region has very few reps, suggesting limited presence or lower priority. This distribution may align with current revenue sources but expanding in the International or Midwest regions could offer new growth opportunities.

Q2: Look into revenue streams

The products that Parch offers include standard, glossy, and poster paper. The revenue streams are investigated by analyzing how much each product contributes to total quantity sold and total sales. The difference in those two proportions can be explained by the average price for each product. Standard paper stands out as the sole product that contributes to over 52.7% of total product sold and about 40% of total sales. They also have the lowest price among all 3 products.

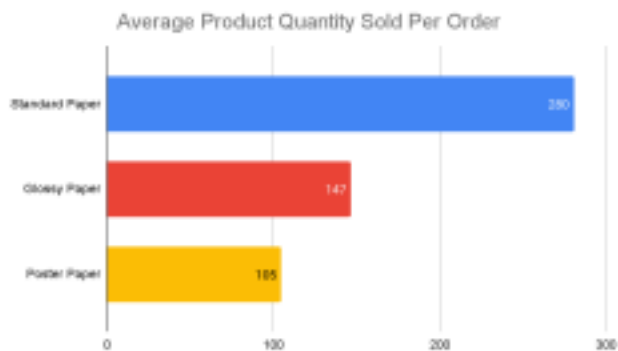
select avg(standard_qty) as st

	0-9 stavg	0-9 glossavg	0-9 posavg
1	280.4320023148	146.6685474537	104.6941550926

Results 1 X				
select sum(standard_amt_usd) Enter a SQL expression to filter results (k				
Grid	0-9 ?column?	0-9 ?column?	0-9 ?column?	
1	4.99	7.49	8.12	
txt				



Total Product Quantity Sold: Standard Paper dominates sales at 52.7%, followed by Glossy Paper at 27.6% and Poster Paper at 19.7%, indicating a strong preference for Standard Paper among customers.



Average Product Quantity Sold Per Order: Standard Paper has the highest average quantity per order at 280 units, suggesting it is purchased in larger volumes than Glossy (147) and Poster Paper (105).



Average Price Per Product: Poster Paper has the highest average price at \$8.12, followed by Glossy Paper at \$7.49, with Standard Paper being the most affordable at \$4.99.

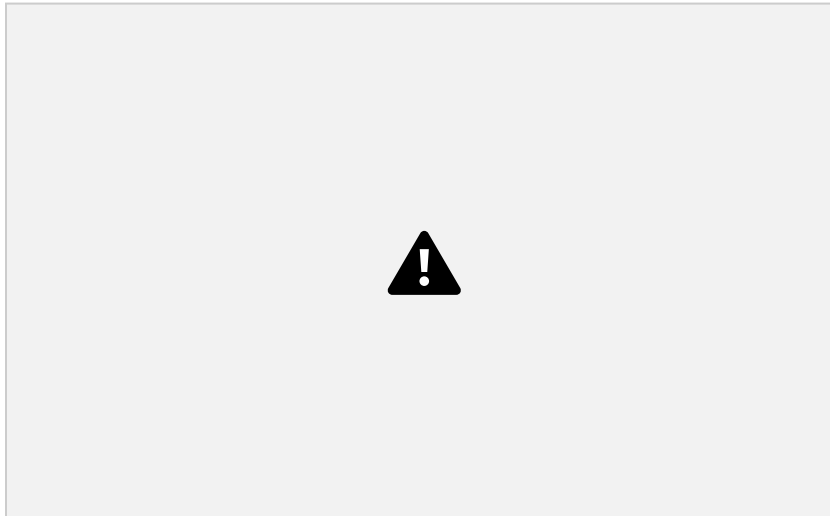
Q3: Is the business growing?

Revenue growth over time

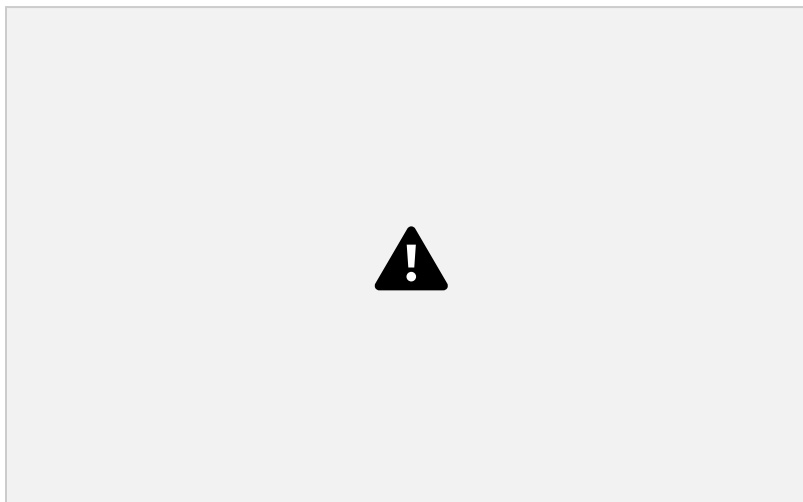
From this analysis, we observed relatively stable revenue until mid-2015, followed by steady growth, with a sharper increase beginning in 2016. This trend suggests that the company experienced successful revenue expansion, especially in the last year.



Revenue Growth Over Time: Revenue shows steady growth from 2014 to late 2016, with a sharp increase towards the end of 2016, followed by a sudden drop at the beginning of 2017.



Sales Repartition Over the Years: Standard Paper consistently leads in sales from 2014 to 2016, with all product types experiencing growth, especially significant for Standard and Glossy Paper in 2016.



Monthly Sales Evolution by Product Type: Sales for all product types show an upward trend, with Standard Paper leading and experiencing a sharp increase starting in 2016, followed by a drop at the beginning of 2017.



Customer Growth & Retention Over Time: Total and repeat customer numbers steadily increase over time, with a strong rise in 2016, indicating growing customer loyalty and acquisition until a sudden decline in early 2017.

Q4: Could we increase sales efficiency?

a) We obtain a table where each sales rep is directly associated with a region and the number of accounts they are responsible for. As shown in the table, one sales rep is assigned to as many as 15 accounts, while some are responsible for only 3 accounts. This indicates an uneven allocation of sales representatives to accounts. However, some accounts have higher sales than others. By joining the orders table and summing the total amount of paper sold by account ID, we can link each sales rep to both the number of accounts they manage, and the total sales associated with those accounts. Visualizing this data reveals a positive correlation between the number of accounts and the total sales amount. Considering reassigning some sales reps from the Northeast to other regions may be beneficial.



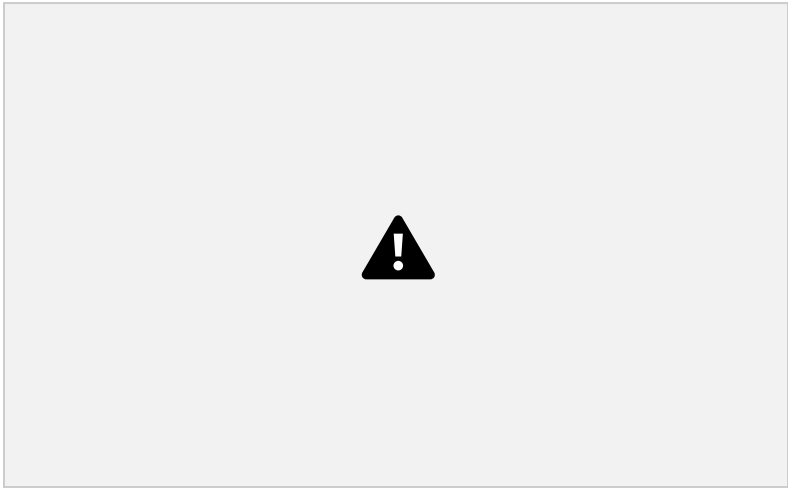


b) Since Dunder Mifflin has added North, South, and International as new regions, it's essential to allocate sales reps to these areas effectively. Running the query below we can find the total number of sales reps associated with each region, which would give us an idea of how to reallocate the sales reps to each region. For example, region 1 (Northeast) has the most sales reps so reallocating some of them to region 6 or 7 might be a good idea.



Running the query below allows us to find the total sales for each region, which can inform resource reallocation decisions. As shown, the Northeast region has the highest sales volume but also the most sales reps. For instance, the Northeast has nearly twice the number of sales reps as the Southeast, yet its sales amount does not reflect a proportional increase. Therefore, reallocating some sales reps to other regions remain a logical step.





Number of Sales Reps in Northeast: 41.2%

Number of Sales Reps in Southeast: 19.6%



Total Sales in Northeast: 33.5%

Total Sales in Southeast: 28.2%



Q5: Marketing Strategy

a) We used text pattern matching in SQL to identify industries based on customer names. Examples of potential industries included “Food,” “Health,” “Education,” and “Retail.” We then identified the top customers in each industry and the least-used channel for web events in each old region to deactivate. Our queries returned matches for the Food, Health, Retail/Consumer Goods, and Marketing/Advertising industries. However, there were no results for Education, Publishing, or Printing. Based on these findings, we decided to focus our marketing content on the Food and Health industries.



Food industry



Health industry



Retail/ Consumer Goods industry



Marketing/ Advertising industry returns 1 result



Education industry returns 0 result

Publishing/ Printing returns 0 result

Marketing Strategies for Food and Health Industries

Food Industry: Educational Content on Food-Safe Materials

- Content Theme: “Ensure Safety with FDA-Compliant Packaging Materials” ● Content Format: Blog posts focused on the importance of food-safe materials in packaging. ● Key messages: Highlight the importance of using FDA-compliant, food-safe packaging to avoid contamination. Provided detailed information on how Parch and Posey’s paper products meet regulatory standards.
- Distribution channels: Email campaigns to existing food industry clients, LinkedIn posts, and downloadable resources on the website.

Health Industry: Eco-Friendly Medicine Packaging

- Content Theme: “Safe, Sustainable Solutions for Medicine Packaging”
- Content Format: Brochure, short video to showcase eco-friendly and demonstrate product safety and eco-friendly materials
- Key messages: Eco-friendly packaging that ensures patient safety and environmental

responsibility that meet health industry standards for safe medicine storage and tailor different medication types and storage needs.



- Distribution channels: Target pharmaceutical and healthcare providers, healthcare trade shows, healthcare fair, and LinkedIn posts.

b) Based on the query results, the least-used channels for web events in each region are Midwest (Banner), Northeast (Twitter), Southeast (Twitter), West (Banner). We use this information to temporarily deactivate the least-used channels in each of these regions to free up resources for focusing on the new regions. This decision is based on the

channels being the least active for web events in each respective region, which may suggest lower engagement or relevance for customers in those areas. This allows us to allocate resources more effectively toward channels that perform better in existing regions or focus on newly added regions.

Least used channels are banner (Midwest), twitter (Northeast), twitter (Southeast), banner (West)

V. Recommendation

- **Overall Recommendation:** We recommend acquiring Parch & Posey due to its consistent growth and stability over the years. This strong growth trajectory highlights its profitability

potential, making it a valuable addition to Dunder Mifflin's portfolio. By acquiring Parch & Posey, Dunder Mifflin can strengthen its market position and secure a reliable new revenue stream.

- **Key Considerations:** It's important to clearly define the target market for Parch & Posey's products to maximize acquisition benefits. Identifying specific customer segments based on demographics, geographic preferences, and unique needs will allow Dunder Mifflin to tailor its strategies effectively. This focused approach can enhance customer engagement and increase market share.
- **Areas for Improvement:** Optimizing resource allocation across regions can improve sales efficiency. By aligning sales rep distribution with customer demand in each region, Dunder Mifflin can ensure that representatives are assigned where they are most needed. This reallocation will help improve customer satisfaction and drive sales growth, ensuring that resources are utilized effectively across all regions.